Business Mathematics – LastMinutePDF Premium Rapid Revision Guide

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Premium Rapid Revision Guide

Percentages & Ratio

KEY FORMULAS

Percentage

% = (Part ÷ Whole) × 100

Successive % Change

Final = Initial × (1 + a/100) × (1 + b/100)

Ratio & Proportion

If a:b = x:y, then a/b = x/y

BUSINESS USES

- Discount calculations on MRP
- Markup pricing on cost price

SOLVED EXAMPLES

Ex 1 — Basic Percentage

A student scores 72 out of 90.

% = (72 ÷ 90) × 100 = 80%

Ex 2 — Successive Change

Price rises 10%, then falls 5%.

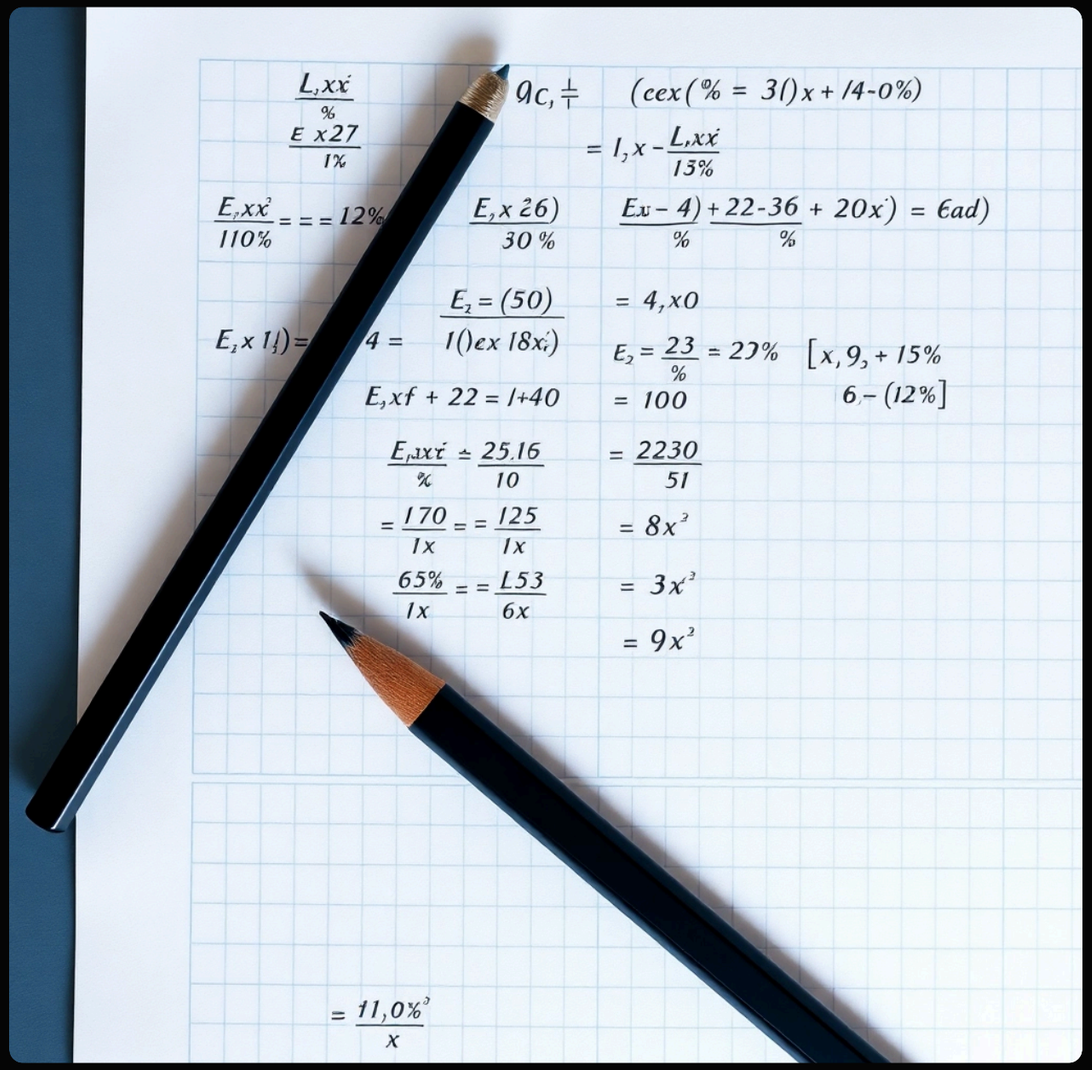
Final = 100 × 1.10 × 0.95 = ₹104.50

Net change = +4.5%

Ex 3 — Ratio & Proportion

If 4:6 = x:9, find x.

4/6 = x/9 → x = (4 × 9) ÷ 6 = 6



Simple & Compound Interest

High Priority · DeepDive

Simple Interest (SI)
SI = (P × R × T) / 100
A = P + SI
Where P = Principal, R = Rate %, T = Time (years)
Compound Interest (CI)
A = P(1 + R/100)^T
CI = A − P
Half-yearly: A = P(1 + R/200)^2T
SI vs CI

- SI: interest on principal only
- CI: interest on principal + accrued interest
- CI > SI for same P, R, T (T > 1 yr)
- Equal for T = 1 year (annual compounding)

Solved Examples
Ex 1 (SI): P=1000, R=5%, T=3 yrs
SI = (1000×5×3)/100 = ₹150
Ex 2 (SI): P=5000, R=8%, T=2 yrs
SI = (5000×8×2)/100 = ₹800
Ex 3 (CI Annual): P=1000, R=10%, T=2
A = $1000(1.1)^2$ = 1210 → CI = ₹210
Ex 4 (CI Half-Yearly): P=1000, R=10%, T=2
A = $1000(1.05)^4$ = 1215.5 → CI = ₹215.5
Ex 5 (SI vs CI): P=2000, R=10%, T=2
SI = ₹400 · CI = $2000(1.1)^2 - 2000$ = ₹420
Difference = ₹20

TIME VALUE OF MONEY

Key Formulas · $PV = FV / (1+r)^n$ · $FV = PV(1+r)^n$ · PV of Annuity $= PMT \times [1 - (1+r)^{(-n)}] / r$

EMI (Equated Monthly Instalment) spreads a loan into equal monthly payments covering principal + interest. Use the annuity PV formula with r = monthly rate and n = total months to derive the EMI amount.

SOLVED EXAMPLE 1

Future Value of a Lump Sum

You invest ₹50,000 today at 8% p.a. for 3 years. What is the future value?

$FV = PV(1 + r)^n = 50{,}000 \times (1.08)^3$

$= 50{,}000 \times 1.2597 =$ ₹62,986

The investment grows by ₹12,986 due to compounding over 3 years.

SOLVED EXAMPLE 2

EMI on a Personal Loan

Loan = ₹1,00,000, annual rate 12% (1% monthly), tenure 12 months. Find the EMI.

$EMI = P \times r / [1 - (1+r)^{(-n)}]$

$= 1{,}00{,}000 \times 0.01 / [1 - (1.01)^{(-12)}]$

$= 1{,}000 / 0.1126 =$ ₹8,885 / month

Total repayment = ₹1,06,620 → interest cost = ₹6,620.

Break-Even & Cost-Volume Logic

CVP Analysis · Core Formulas & Examples

Key Formulas

Break-Even (Units)
= Fixed Cost ÷ (SP − Variable Cost)

Contribution
= Selling Price (SP) − Variable Cost (VC)

Profit
= (SP − VC) × Quantity − Fixed Cost

Margin of Safety
= Actual Sales − Break-Even Sales

Break-Even Graph Logic

The graph plots **Total Revenue** and **Total Cost** against output (units). They intersect at the **break-even point**. Below it: loss zone. Above it: profit zone. The gap between actual sales and

Example 1

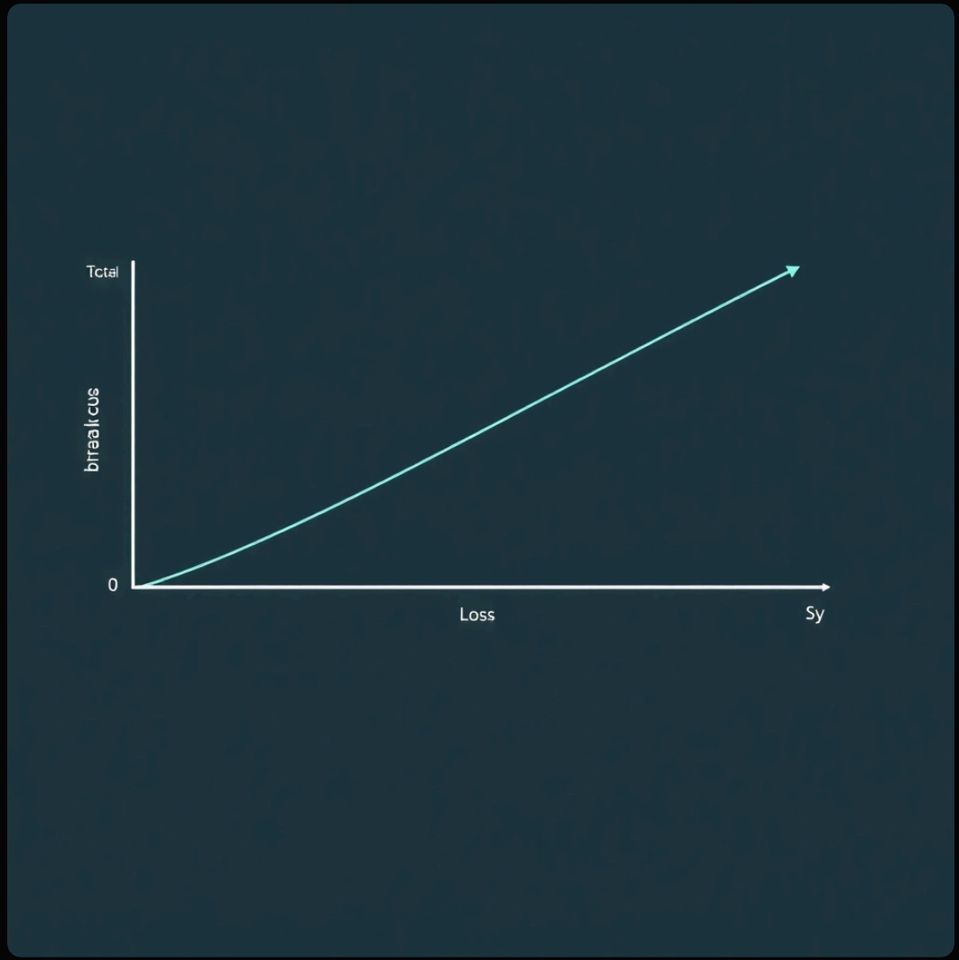
FC = ₹60,000 | SP = ₹50 | VC = ₹30

Contribution = 50 − 30 = **₹20**
BE Units = 60,000 ÷ 20 = **3,000 units**
If Q = 4,000:
Profit = 20 × 4,000 − 60,000 = **₹20,000**
Margin of Safety = 4,000 − 3,000 = **1,000 units**

Example 2

FC = ₹1,20,000 | SP = ₹80 | VC = ₹50

Contribution = 80 − 50 = **₹30**
BE Units = 1,20,000 ÷ 30 = **4,000 units**
If Q = 5,500:
Profit = 30 × 5,500 − 1,20,000 = **₹45,000**
Margin of Safety = 5,500 − 4,000 = **1,500 units**



Arithmetic & Geometric Progressions

ARITHMETIC PROGRESSION (AP)

nth term: $a_n = a + (n - 1)d$
Sum of n terms: $S_n = n/2 \times [2a + (n - 1)d]$
Where: a = first term, d = common difference, n = number of terms

GEOMETRIC PROGRESSION (GP)

nth term: $a_n = a \times r^{(n-1)}$
Sum of n terms: $S_n = a(1 - r^n) / (1 - r), r \neq 1$
Where: a = first term, r = common ratio, n = number of terms

BUSINESS APPLICATIONS

- **AP:** Equal instalment repayments, salary increments by fixed amount
- **GP:** Compound growth rates, population growth, depreciation by fixed %
- If differences are **constant** → AP. If ratios are **constant** → GP.
- For GP with r > 1: growth series. For r < 1: decay/depreciation.
- Sum formula changes to $S = na$ when r = 1.
- Always identify a, d/r and n before substituting.

SOLVED EXAMPLES

1. **AP:** A firm repays a loan in instalments: ₹5,000; ₹5,500; ₹6,000...
   Find the 10th instalment → a=5000, d=500
   $a_{10} = 5000 + 9 \times 500 = ₹9,500$

2. **GP:** Sales grow at 20% p.a. from ₹1,00,000.
   Sales in year 5 → a=1,00,000, r=1.2
   $a_5 = 1,00,000 \times 1.2^4 = ₹2,07,360$

3. **GP Sum:** Total revenue over 4 years if first year = ₹50,000, r = 1.1
   $S_4 = 50,000 \times (1.1^4 - 1) / 0.1$
   $S_4 = ₹2,32,050$

Logarithms & Exponents

Log Rules

Core Log Rules

Product Rule
$\log(ab) = \log a + \log b$

Quotient Rule
$\log(a/b) = \log a - \log b$

Power Rule
$\log(a^n) = n \cdot \log a$

Applications

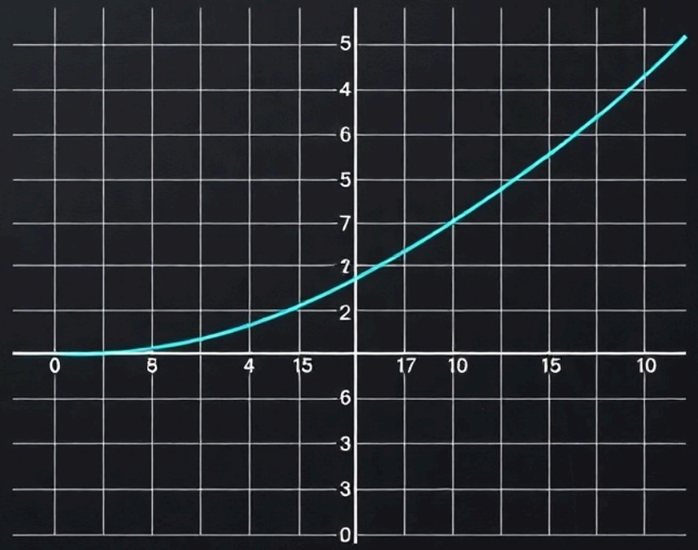
- Solving for time in compound interest formulas
- Calculating growth rates from exponential data

Solved Example 1 — Compound Interest
How many years to double at 8% p.a. compounded annually?

$A = P(1 + r)^n \rightarrow 2P = P(1.08)^n$
$\log 2 = n \cdot \log 1.08$
$n = \log 2 \div \log 1.08$
$n = 0.3010 \div 0.0334$
$n \approx 9.01$ years

Solved Example 2 — Growth Rate
A city grows from 50,000 to 80,000 in 5 years. Find annual growth rate.

$80{,}000 = 50{,}000 \cdot (1 + r)^5$
$\log(1.6) = 5 \cdot \log(1 + r)$
$\log(1 + r) = 0.2041 \div 5 = 0.04082$
$1 + r = \text{antilog}(0.04082) \approx 1.0986$
$r \approx 9.86\%$ per year



Most Repeated University Questions

45 exam-ready questions across three parts — modelled on real BBA university paper patterns.

Part A – Long Answer (15 Qs)

1. Calculate CI vs SI on ₹50,000 at 10% p.a. for 3 years.
2. Find NPV of a 5-year project at 12% discount rate.
3. A trader marks up 40% then discounts 20% — find net profit %.
4. Solve: 8th term of an AP where a=5, d=3.
5. Determine break-even units when FC=₹1,20,000, SP=₹80, VC=₹50.

Part B – Short Numerical (20 Qs)

1. Express 3:5 as a percentage.
2. Find SP if CP=₹200 and profit=15%.
3. SI on ₹10,000 at 8% for 2.5 years.
4. $\log_2 64 = ?$
5. Find the 10th term of GP: 2, 6, 18...

Part C – Mixed Application (10 Qs)

1. A sum doubles in 5 yrs (SI). At same rate, find CI for 3 yrs.
2. Using log tables, solve: 2^x = 500.
3. Combine ratio, % and discount: Find final price after 3 markdowns.
4. A GP sum of 6 terms = 315; find r if a = 5.
5. Break-even + margin of safety combined problem.

How to Attempt These Questions

Start with the formula. Write what is given vs what is asked. Show step-by-step working — marks are awarded for method. Round to 2 decimal places unless stated. Attempt all parts even partially.

Topics Covered Across All Parts

Percentages · Ratio · Profit & Loss · Discounts · SI & CI · Time Value of Money · Break-even Analysis · AP & GP · Logarithms & Exponents

Quick tricks & calculation hacks

Multiplier trick

Convert % to a multiplier: +20% → ×1.2, −15% → ×0.85. Chain multiplications instead of step-by-step additions.

Successive gain/loss

Net effect = a + b + (a×b)/100. For +20% then −20%: 20 − 20 − 4 = −4%. No multi-step working needed.

CI mental approximation

CI ≈ SI + SI×r/100 for 2 years. For quick checks, use $(1+r)^n \approx 1 + nr + n(n-1)r^2/2$ for small r.

Break-even fast solve

BEP (units) = Fixed Cost ÷ Contribution per unit. Memorise this ratio — no equation setup required in exam.

Calculator strategy

Use memory (M+/MR) to store intermediate values. For compound interest, use the power key: 1.08 ^ 5 directly.

Rapid Revision Formula Sheet

01 Interest

SI = $P \times R \times T / 100$

CI = $P(1 + R/100)^n - P$

Amount (CI) = $P(1 + R/100)^n$

02 Percentages

% of X = (Part / Whole) × 100

% Change = (New − Old) / Old × 100

x% of y = y% of x

03 Profit, Loss & Discount

Profit% = (Profit / CP) × 100 | Loss% = (Loss / CP) × 100

SP = CP × (1 ± %/100)

Discount = MP − SP | Discount% = (Discount / MP) × 100

04 Present Value & Future Value

FV = $PV \times (1 + r)^n$

PV = $FV / (1 + r)^n$

Annuity FV = $PMT \times [(1+r)^n - 1] / r$

05 Break-Even (BEP)

BEP (units) = Fixed Costs / (SP − VC per unit)

BEP (sales) = Fixed Costs / Contribution Margin Ratio

CM Ratio = (SP − VC) / SP

06 AP & GP

AP: n^{th} term = $a + (n-1)d$ | Sum = $n/2 \times [2a + (n-1)d]$

GP: n^{th} term = ar^{n-1} | Sum = $a(r^n - 1) / (r - 1)$

$GP_\infty = a / (1 - r)$, where $|r| < 1$